DEFINING THE BEVERAGE

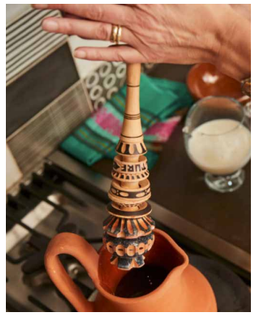
1

I started the design process by chosing a spesific beverage. I did not iclude AI in this part of the process.

Initial Research (Empathize)

- Investigated the traditional preparation of Mexican champurrado.
- Identified that preparation involves **continuous stirring with a molinillo**, heating in a pot, and visible foam formation.
- Recognized the drink as a **cultural ritual rather than a purely functional cooking task**.
- Early design question: *Should the product replace the traditional process or support it?*

Design Question: How to assist without replacing embodied ritual?



EXAMINING THE RITUAL

2

User & Context Definition (Define)

- Defined the **primary user** as a café owner interested in presenting cultural beverages.
- Identified the café as a **semi-open environment where preparation is visible to customers**.

Key insight:

Champurrado preparation is a **multisensory ritual involving steam, sound, and physical gesture**.

Identified tensions:

- Ritual vs fast café environment
- Sensory ritual vs mechanical café atmospher

Prompt 1: What could be the user considerations of this drink (like primary users and its context of use), if I were to design an industrial product for preparing Champurrado?

Prompt 2: How about some early assumptions and design direction you would form while designing this product?

CLARIFYING THE EMPHATIZATION PHASE

3

Prompt 3: Summarize and explain the key research findings of this research.

Understand hot beverage preparation as a lived ritual, use this as a guideline:
a) time pressure, habits, culture, domestic context,
b) existing devices and their interaction logic,
c) sensory experience (steam, sound, light, tactility),
d) safety and ergonomics realities

Insight: The ritual demands attention in time-constrained contexts.

Insight: Removing visible preparation and containing it into a closed machine counterpart, risks losing the cultural connection and value of the "ritual".

Insight: Knowledge of readiness is tactile and experience based rather than precise or digital.

Insight: Tools must integrate into compact, modern kitchens while respecting and conserving the traditional form and culture.

Insight: Removing visible preparation and containing it into a closed machine counterpart, risks losing the cultural connection and value of the "ritual".

AI as Expander

AI as Clarifier

CLARIFYING THE USER AND THE ENVIRONMENT

4

Prompt 4: The primary user is a cafe owner that is interested in different cultures and presenting them to his customers in the cafe. Give me a clear user definition, and a single paragraph of a user scenario for him, you can add what you think is going to be important for me to consider in the design process: think of things like where its going to be used, who is going to use it and how it's going to be used. Also take in consideration the ritual of the preparation process. (I also defined the use scneario to the AI detailed here)

Refined User Definition

An independent café owner who likes to include global beverages in his menu in a semi-open bar setting. He values cultural authenticity, sensory atmosphere, and visible preparation, while operating under commercial time pressure and shared staff usage.

Clear Use Scenario

In a semi-open café environment, the owner prepares champurrado in front of customers as part of a seasonal cultural offering.

1) After the drink is ordered the device is brought to the customers table with the drink already inside.

2) Its starts heating and the customers can observe the readiness of the drink by the abundance of steam.

3) Than the funnel part of the product is removed by holding the tip of the cup and the second part of the drink (masa powder for thickhening) is added to the device.

4) Then the barista proceeds to whisk the drink with the traditional molinillo.

5) The funnel is placed on top again and the drink left to heat.

6) When the drink is ready the funnel is once again removed and the bottom cup is held by the handles and is placed in front of the customer as a drinking cup.

5

How do we modernize without flattening cultural meaning?

DESIGN OPPORTUNITIES AND POSSIBLE PAIN POINTS

Prompt 5: What could be some opportunity areas or pain points regarding what we found as the preparation steps and the overall experience of the product. Keep in mind my user and scenario as well as the environment?

Design Opportunities

- **Ritual visibility:** Make steam, whisking, and transformation clearly visible in cafés.
- **Time pressure:** Structure the ritual so it works during fast service.
- **Ergonomics:** Reduce whisking strain while preserving the traditional gesture.
- **Sensory cues:** Emphasize steam, sound, aroma, and texture.

Pain Points

- **Ritual vs café speed:** Slow preparation conflicts with fast service.
- **Physical fatigue:** Continuous molinillo whisking strains the wrist.
- **Burning risk:** Thick mixture easily burns at the bottom.
- **Low visibility:** Preparation can disappear among café machines.
- **Cleaning difficulty:** Masa residue sticks to vessel surfaces.

Design Question: How can the system teach the ritual implicitly?

AI as Critique
AI as Re-Framer
AI as Expander
AI as Clarifier

DEFINING THE DIRECTION

6

7

CONCEPT EXPLORATION

Prompt 4: Which direction would still matter if technology became perfect?

Design Direction

Even if technology solved all functional problems—perfect heat control, no burning, effortless whisking, consistent texture, and easy cleaning—the core tension would remain: the slow ritual of champurrado risks disappearing within the fast, mechanical café environment.

Therefore, the project focuses on **preserving and amplifying the cultural ritual of preparation**, rather than optimizing technical efficiency.

Selected Top Tensions

- **Ritual vs. Time Pressure** – Slow preparation within fast café service.
- **Sensory Ritual vs. Mechanical Café Atmosphere** – Steam, sound, and gesture risk being lost among café machinery.

Problem Framing

How might we support the visibility and embodied meaning of champurrado preparation in a fast café environment without automating the ritual?

Direction Decision

Focus on **amplifying ritual presence**, not technical optimization.

Rejected Directions

- Full automation of preparation
- Pure heat-control solutions
- Staff-consistency systems
- Large industrial production devices

These approaches solve functional issues but **remove or weaken the ritual experience**.

Concept Exploration

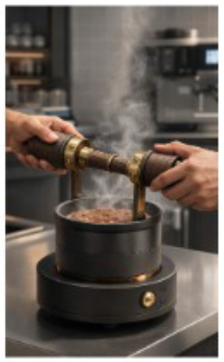
Three design concepts were explored to address the tension between the **slow ritual of champurrado preparation and the fast-paced café environment**.

Concept 1 – Ritual Amplifier
This concept focuses on making the preparation visible and performative. The design elevates and frames the whisking process, highlighting steam, motion, and sound so the ritual becomes a readable experience within the café.

Concept 2 – Co-Ritual System
This concept transforms preparation into a shared activity. Two users whisk the drink together, turning the process into a social interaction that emphasizes participation and collective experience.

Concept 3 – Anchored Ritual Vessel
A dedicated bowl with an integrated heating base that preserves the traditional preparation method. Instead of modifying the ritual, it anchors the practice spatially and gives it a clear presence within the café.

These concepts explore different ways of preserving the ritual: **amplifying visibility, encouraging social participation, or anchoring the tradition through a dedicated object.**



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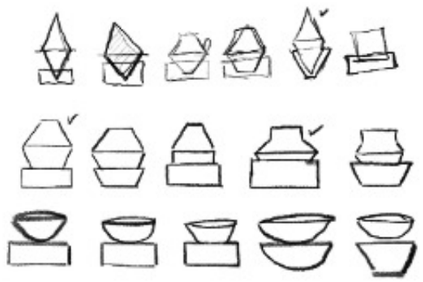
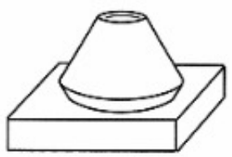
FINDING THE FORM BY TESTING AND PROTOTYPING





Prompt 5 : Test this product regarding it's use, ergonomics, hazards, possible problems and opportunities.

Product Evaluation Summary

The product was evaluated in terms of **use, ergonomics, hazards, functional issues, and opportunities** within a café table-side preparation scenario.

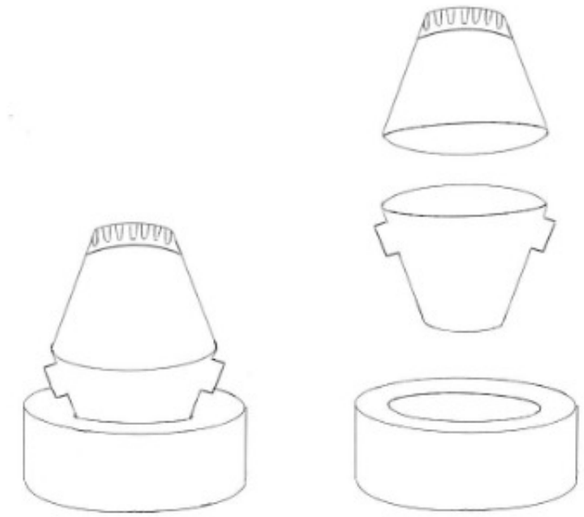
Ergonomics
The main concern is handling the hot funnel during removal. Possible improvements include thicker grip areas, heat-resistant materials, and increasing base stability to prevent movement during whisking.

Hazards
Potential risks include heat from the chimney, steam exposure, and splashing during whisking. These can be reduced through improved grip geometry, controlled steam direction, and a thicker vessel rim.

Functional Issues
Cleaning may be difficult due to sticky champurrado residue, and the funnel components require precise alignment. Smooth interior surfaces and simple connection features can improve usability.

Opportunities
The design creates a clear ritual sequence—heating, steam stage, whisking, and serving—which enhances the performative experience. The funnel can also function as a molinillo holder or resting tool.

Accepted Improvements
Grip geometry on the funnel, a rubber base ring for stability, alignment features between parts, and handles on the vessel for safe handling were accepted and integrated into the design.



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WHAT DID I NOT THINK OF?

9

REITERATION, IMPROVEMENTS AND FINALIZATION

10

Prompt 6 : After all the decisions I chose, and all the developments we did, expand the horizon of questions I asked. What did I not think of? What could be reframed as another question?

Design Reflection and Reframing

The project explored deeper questions about **ritual, performance, material meaning, sensory experience, and object identity**. Instead of focusing only on how the product functions, the design was reframed to consider how it **stages and communicates the ritual of champurrado preparation**.

The final design highlights two key moments of the ritual:

- **Heating stage** – visible steam rising through the chimney creates anticipation.
- **Whisking stage** – the molinillo produces foam, movement, and sound, forming the main performance.

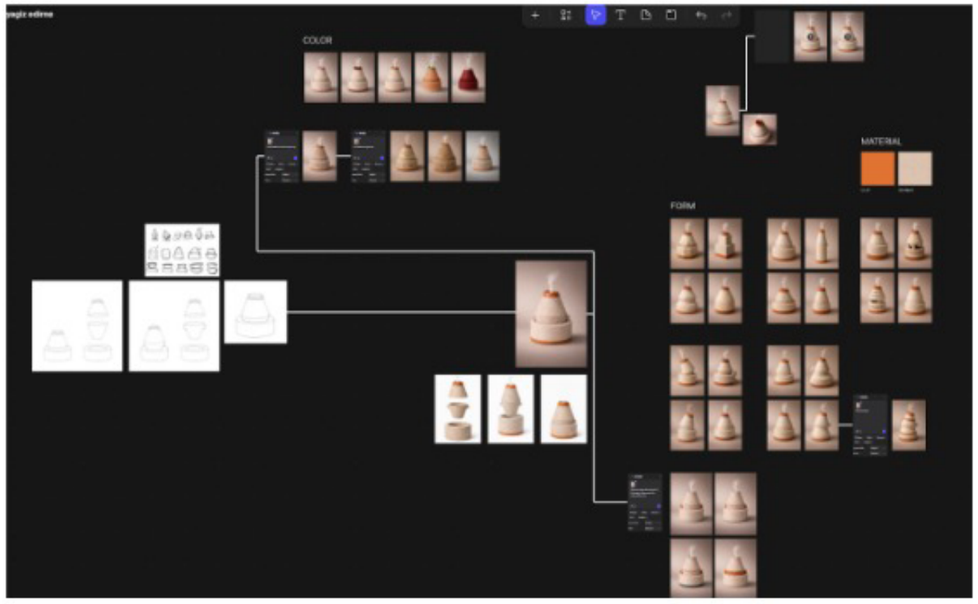
The product reveals important ritual actions such as whisking, steam, and foam while hiding technical components like the heating mechanism.

The object is designed to function simultaneously as **a preparation tool, serving vessel, and ceremonial object**, giving it a strong presence on the table.

Ceramic material connects the design to traditional clay vessels, while modern geometry and silicone details introduce contemporary functionality.

Through steam, sound, heat, aroma, and tactile interaction, the product engages multiple senses and transforms preparation into an **experiential performance**.

Ultimately, the project shifted from designing a device that prepares a drink to designing an **object that amplifies and stages the cultural ritual of champurrado preparation**.



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The product will have no digital interface; in order to not take away from the ritual and keep it authentic. Also the readiness of the drink is understood by visual cues (steam and sound), so a digital feedback is unnecessary.

It will have a heater mechanism inside the base, acting like a stove which will be turned on/off by a button in the back of the product